



The Augmented Library

mcp-calibre: The Complete Usage & Capability Manual

By jumpifequal

Four Pillars of an AI-Readable Library



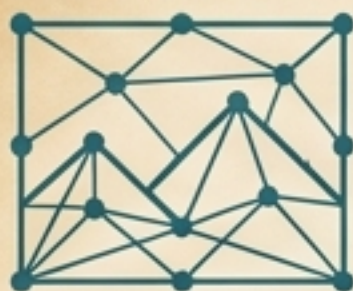
Private & Read-Only

Everything runs locally on your machine. Calibre's databases are opened strictly read-only. There are no write tools, ensuring the AI can never alter your library.



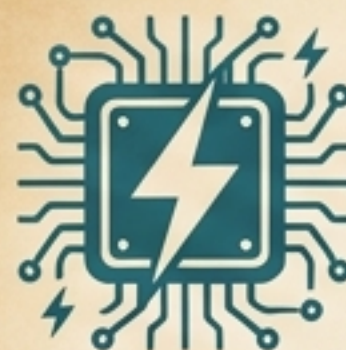
Omnivorous Reading

Effortlessly digests EPUB, PDF, LIT, MOBI, and TXT files. Built-in Tesseract OCR seamlessly translates scanned PDFs into readable text.



Visual & Structural

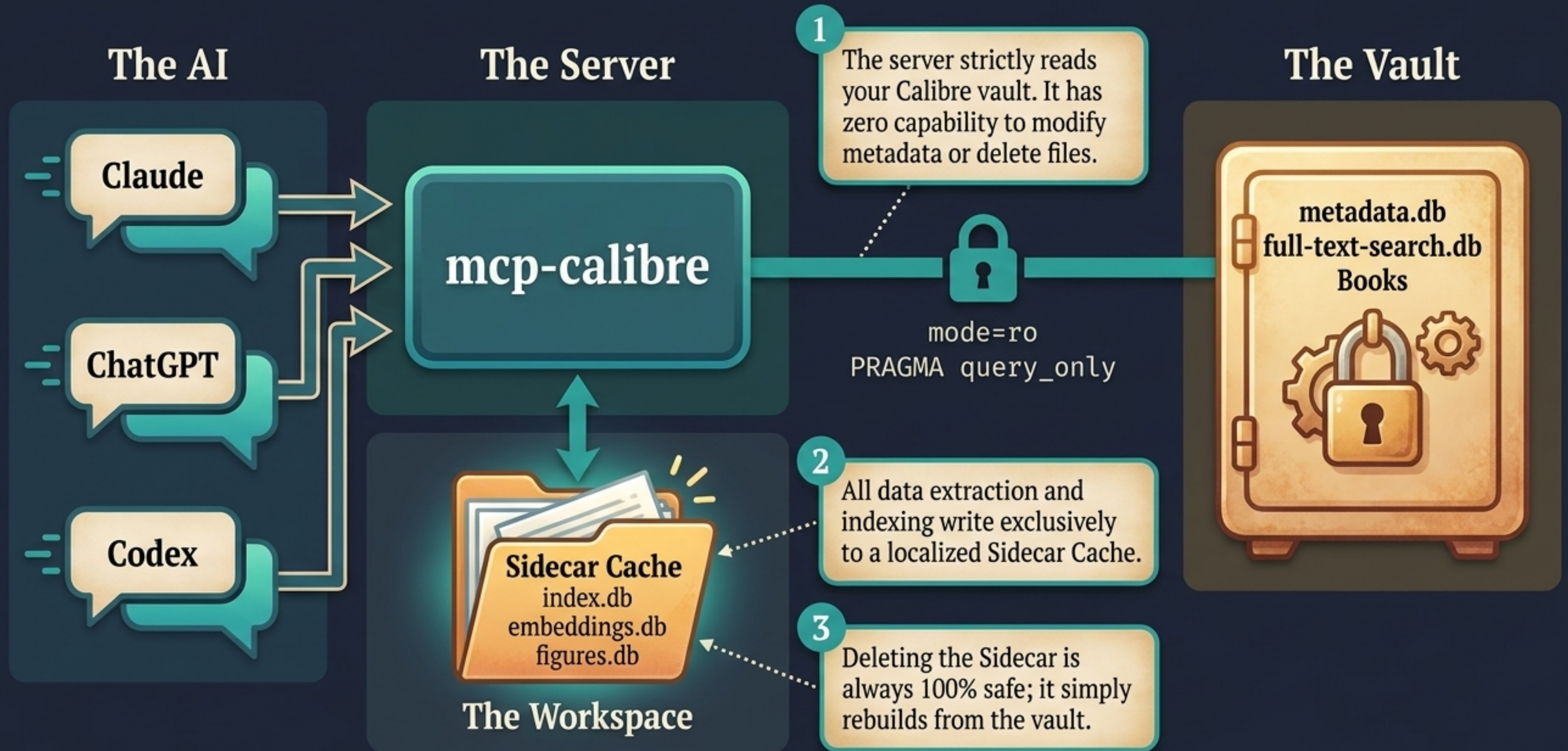
The AI navigates actual book structures via chapter maps and displays inline covers and figures directly in the chat without spending model tokens.



Blazing Fast

Powered by optimized SQLite connections, queries execute in milliseconds, easily handling libraries of thousands of books.

The Architecture of Trust: A Strict Read-Only Guarantee



Deployment Matrix: Getting Connected

Prerequisites: Python 3.10+ (64-bit recommended) and Calibre Full-Text Indexing enabled.

Setup Paths

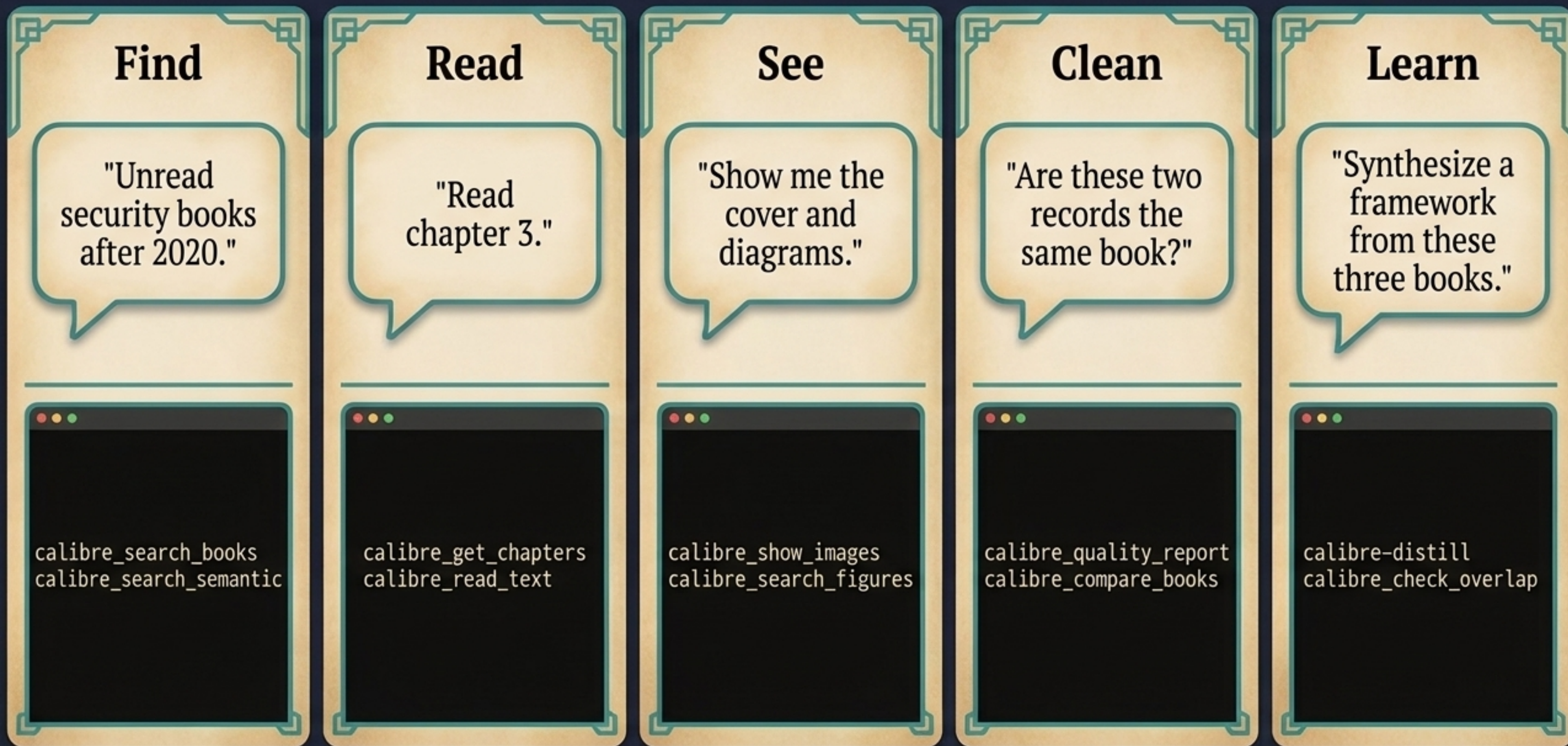
Path	Description
Wizard	GUI-driven, perfect for first installs.
Script	PowerShell (install.ps1), repeatable one-command deployment.
Manual	Full control, required for macOS and Linux.

Supported AI Clients

Client	Connection
Claude Desktop	Native stdio connection (Edit <code>claude_desktop_config.json</code>).
OpenAI Clients (Codex/ChatGPT App)	Native stdio via <code>config.toml</code> .
Claude Code / Shared LAN	Streamable HTTP with mandatory Bearer Token.

Warning: Web ChatGPT and Claude.ai custom connectors are not supported due to required public OAuth exposure of local files.

The Playbook: Mapping 30 Tools to 5 User Intents



Finding Books: The Search Modalities Matrix

LOG [METADATA]: Analyzing query structure and native index.

Metadata Search

Mechanism: Uses native Calibre search syntax.

Speed: Lightning fast (3-8 milliseconds).

Syntax Example:
tag:security and not
#read:true

LOG [FULL-TEXT]: Processing lexical terms and BM25 ranking.

Full-Text Search (Lexical)

Mechanism: Exact word matches, BM25 ranking.

Constraint: Language-dependent (English queries find English words).

Syntax Example:
mode='phrase'
query='lateral movement'

LOG [SEMANTIC]: Computing multilingual vector similarity.

Semantic Search (Meaning)

Mechanism: Multilingual vector matching.

Advantage: Ask in Italian ('infezione'), find English passages ('malware payload'). Chapter-bounded for perfect context.

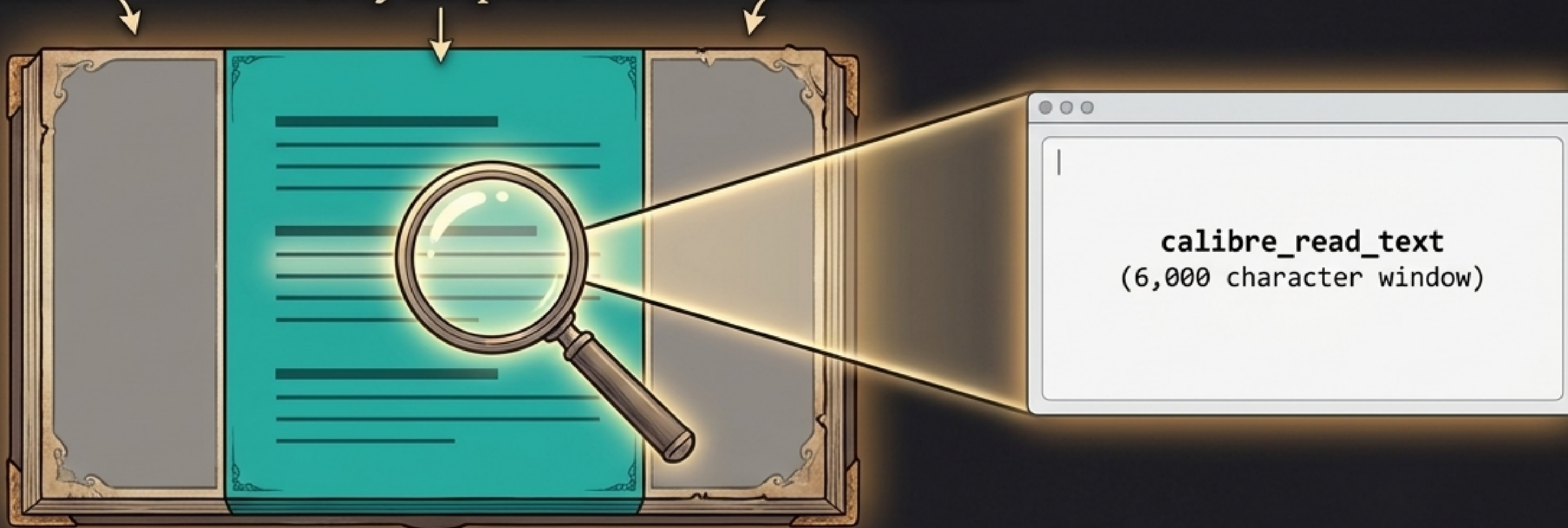
Syntax Example:
mode='hybrid'
query='infezione'

Navigating Structure: How the AI Reads

Front Matter

Body Chapters

Back Matter



The Chapter Map

The `calibre_get_chapters` tool aligns with EPUB Table of Contents or uses heuristic heading detection for PDFs/MOBI to build a structural map, tagging front/back matter to prioritize body content.

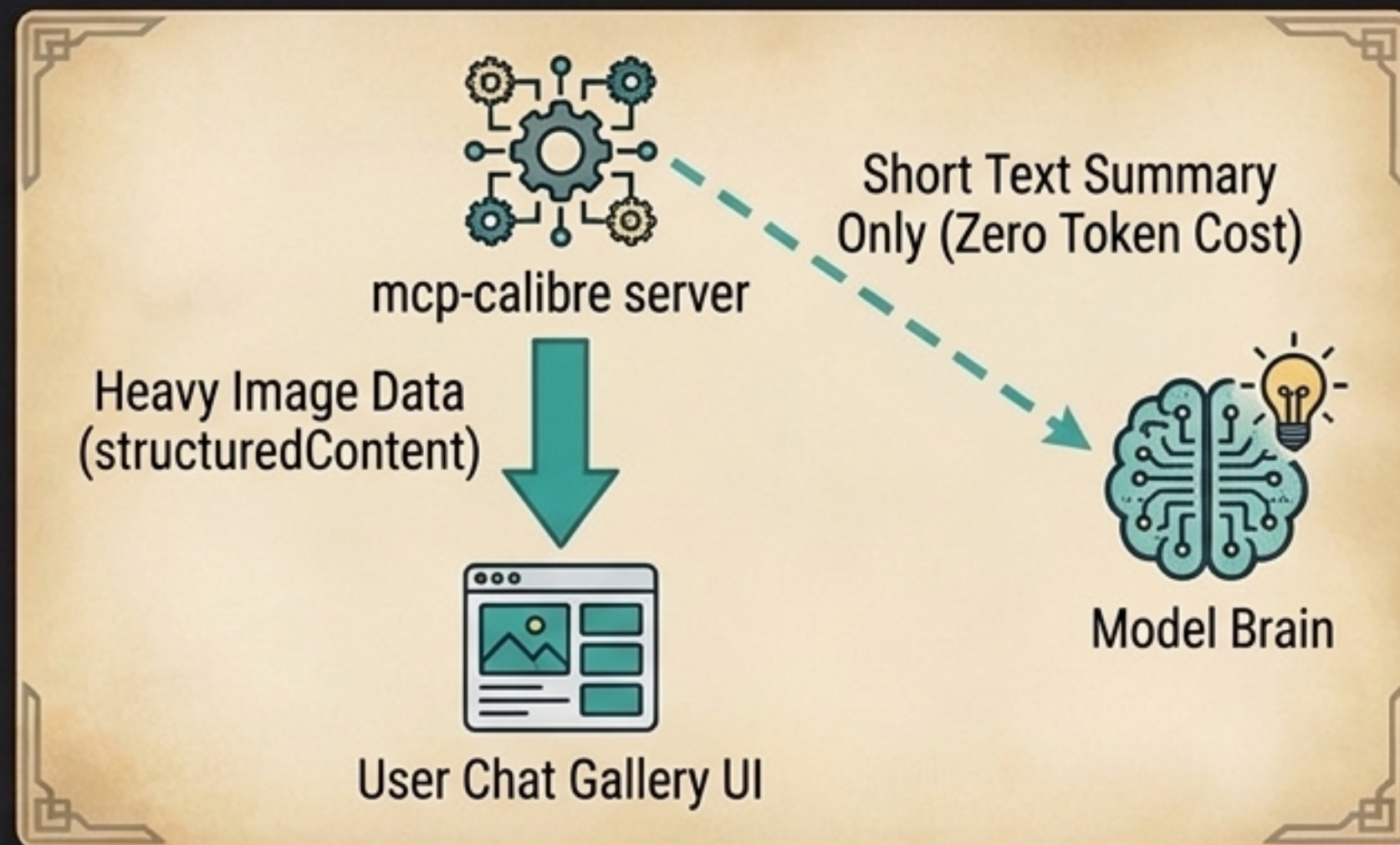
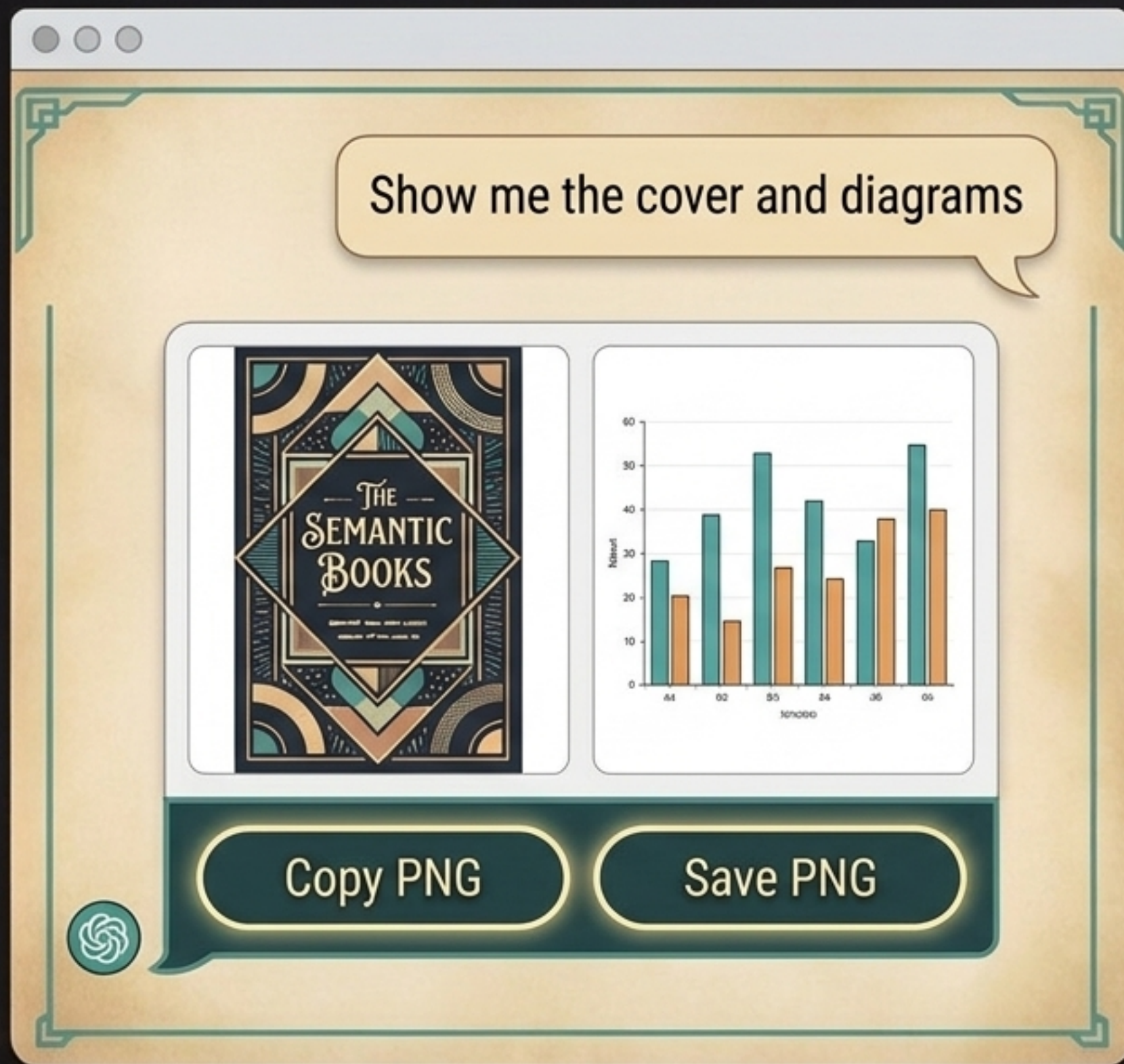
Targeted Extraction

Instead of reading a whole book blindly, the AI requests a specific chapter or a precise 6,000-character window centered on a search hit using `calibre_read_text`.

Format Agnostic

On-demand extraction seamlessly handles EPUB, PyMuPDF, and triggers Calibre's `ebook-convert` for legacy LIT/MOBI formats.

Visualizing the Library at Zero Token Cost



The Feature: ``calibre_show_images`` utilizes the MCP Apps extension (SEP-1865) to render covers and internal book figures directly in your chat interface.

The Magic: Image data travels securely directly to the gallery UI frame. The model never sees the heavy image data, saving immense token costs.

Curation: The Read-Only Robotic Librarian

calibre_quality_report

Diagnostic Alerts



The Quality Report

Audits the entire library for missing authors, invalid ISBNs, raw file-name titles (e.g., BOOK_12_final.pdf), and unsorted author names.

calibre_compare_books

Diagnostic Alerts



Deep Duplicate Detection

Performs strict field-by-field comparisons to find duplicates. It intelligently ignores edition notes but respects subtitles and series numbers, easily flagging likely translations.

	MT	NNF
✓	✓	✓
✓	✓	✓
✓	✓	✓
✓	✓	✗
✓	✓	✗

calibre_find_isbn

Diagnostic Alerts

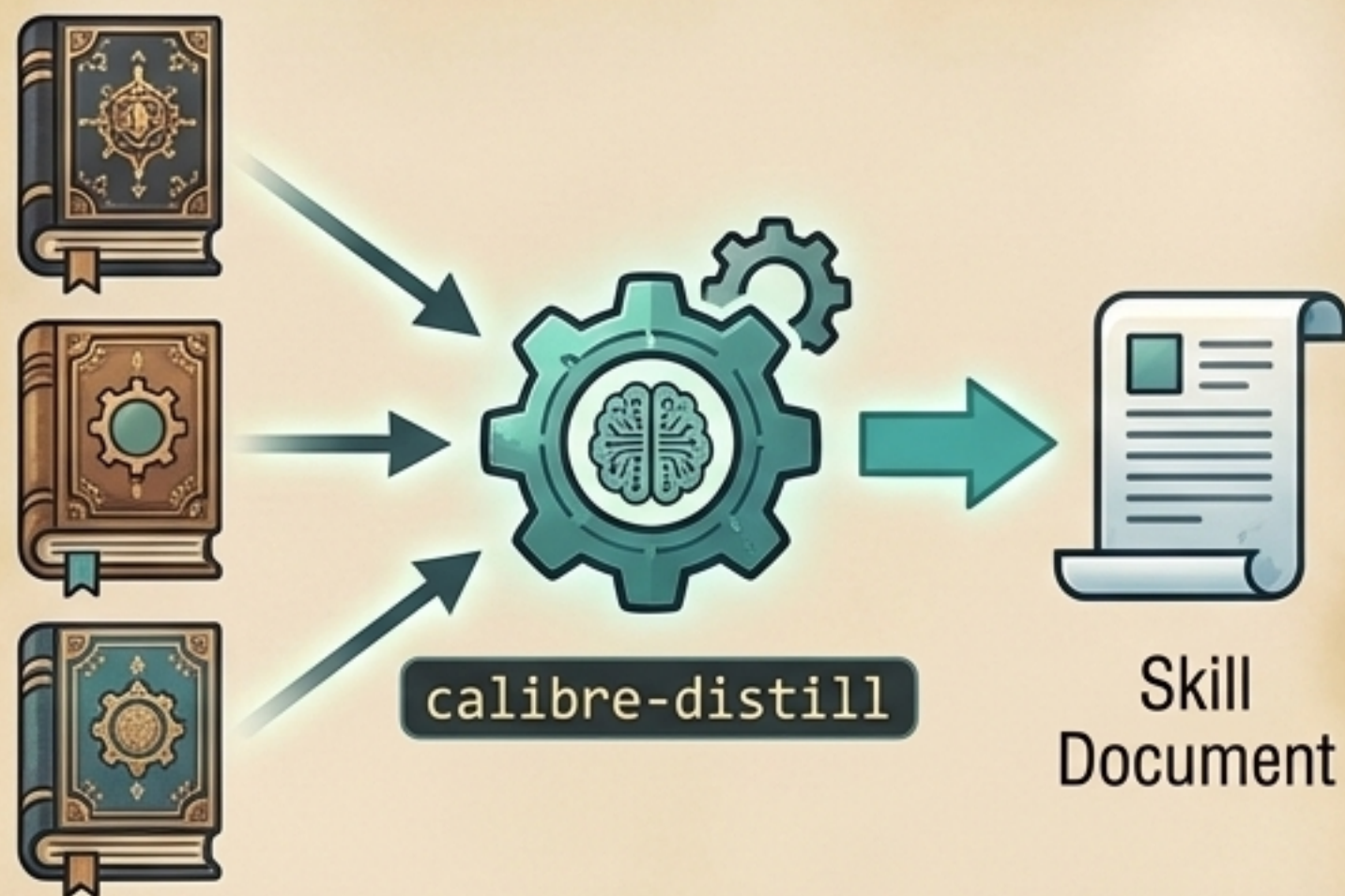


ISBN Text Discovery

Physically scans the copyright pages inside the book's text to discover and checksum-validate the true ISBN when metadata is missing.

Advanced Synthesis & The Mechanical Legal Gate

Companion Agent Skills



Install companion skills to force the AI to read with purpose, paraphrase, and structure knowledge by concepts rather than simply summarizing chapters.

The Mechanical Legal Gate

calibre_check_overlap

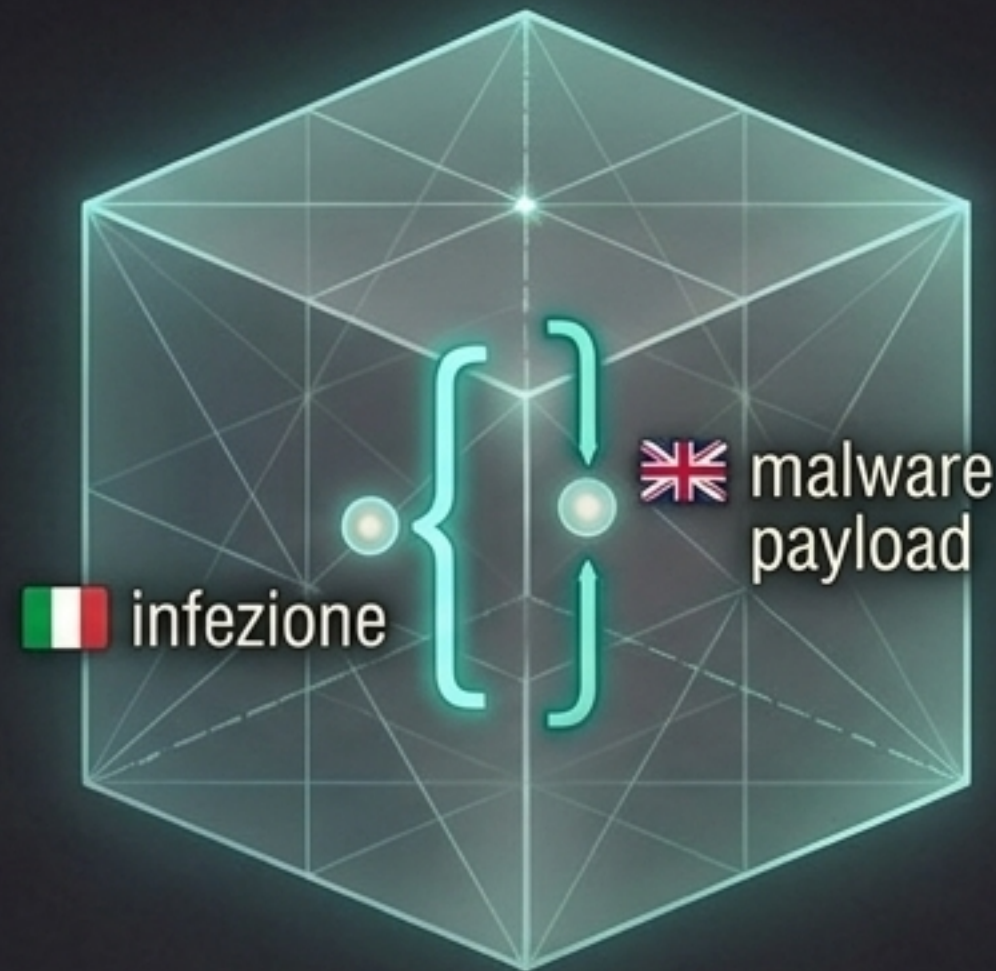
- ✓ **Verbatim Overlap:** Must be $< 3\%$ of 8-word sequences.
- ✓ **Longest Run:** No copied stretches > 20 words.
- ✓ **Quote Budget:** Strictly tracks declared quotes.
- ✓ **Compression:** Notes must be $< 15\%$ of source length.
- ✓ **Attribution:** Every source must be credited by ISBN or Title.

Mechanically guarantees your derived notes do not illegally reproduce the source text.

Under the Hood: The Multilingual Semantic Engine

The Model

Uses **paraphrase-multilingual-MiniLM-L12-v2**, an ONNX model running entirely on your local CPU. Downloaded once, requiring zero network access afterward.



Multilingual Mapping

Sentences across 50+ languages are placed in the same vector space.

Ask in Italian, and the math naturally finds the English passage with the same meaning.

Reciprocal Rank Fusion

Reciprocal Rank Fusion

Combines Int8 vector matches (meaning) with BM25 passage FTS5 (exact keywords) to ensure names and technical IDs aren't lost in the semantic math.

Under the Hood: The Text Extraction Waterfall



Step 1: Calibre FTS

The fastest path. Reads what Calibre has already indexed.



Step 2: Local Sidecar Cache

Incrementally synced `index.db` mapping Calibre's extraction.



Step 3: On-Demand Parsers

Real-time extraction via built-in EPUB parsers and PyMuPDF for clean PDFs.



Step 4: Legacy Conversion

Triggers Calibre's `ebook-convert` for LIT, MOBI, and AZW3 formats.



Step 5: Tesseract OCR

Handled entirely out-of-band via the `--extract-missing` batch command to preserve live chat speed.

Performance Metrics: Engineered for Milliseconds

Based on a synthetic library of 1,500 books running on a single 2.1 GHz CPU core.

Metadata Search

4 - 8 ms

Full-Text Search
(50 books, no snippets)

2.5 - 3.0 ms

Full-Text Search
(10 books, 3 snippets each)

25 - 63 ms

Semantic Hybrid Search

~75 ms

Storage Efficiency

Storage Efficiency

The contentless FTS5 index requires only 148 MB of disk space to index 524 MB of text (just 28% of the total text footprint).

Maintenance & The Graphical Console

The screenshot shows a graphical console window with the following components:

- Tabs:** Profiles, Transport, Maintenance (selected), Logs.
- Settings:**
 - Enable HTTP Supervision: ☒ OFF
 - Server Port: [8080]
 - Enable HTTP Supervision: ☒ ON
 - DPAPI Token: [*****]
 - Enable Server Port: ☐ OFF
 - Server Server Port: None
 - Enable Sync Running: ☐
 - OCR Cormedding: None
- Log Window:**

```
[INFO] Profile 'Default' loaded successfully.
[WARN] HTTP Transport started on port 8080.
[DEBUG] Sync process initiated at 14:32:02.
[INFO] Embedding build queued for 500 documents.
[DEBUG] OCR task running in background thread.
[DEBUG] OCR task running in background thread.
[WARN] Connection retry attempt 2 of 5.
```
- Action Buttons:** --sync, --build-embeddings

- **Profile Management:** Manage all environment variables cleanly in one place without polluting system paths.

- **HTTP Supervision:** Start and monitor the server via Streamable HTTP with secure DPAPI token protection.

- **Maintenance Jobs:** One-click buttons to run sync, embedding, and OCR tasks directly from the GUI.

- **Classified Logs:** Easily debug issues with auto-classified, color-coded output.

Diagnostics & Troubleshooting Cheat Sheet

Symptom	Cause	Fix
'Semantic search skipped' warning.	Semantic engine requires 64-bit Python.	Install 64-bit Python 3.10+ and reinstall.
'OCR needed' error or blank PDFs.	The PDF is a scan without a text layer.	Run <code>calibre_mcp.py --extract-missing</code> to run batch Tesseract OCR.
Italian query finds nothing in an English book.	Full-text search is purely lexical.	Switch to semantic search (<code>calibre_search_semantic</code>) for cross-language matching.
Codex/ChatGPT tool times out on LIT/MOBI files.	Legacy conversions take time.	Raise <code>tool_timeout_sec</code> in your client's <code>config.toml</code> .